# DNS Questions

4 questions | Original English questions and source answers

## Question 1

How does a router behave when configured with the default DNS lookup settings, and a URL is entered on the CLI?

### Options

A. attempts to query a DNS server on the network.

B. prompts the user to specify the desired IP address.

C. continuously attempts to resolve the URL until the command is cancelled.

D. initiates a ping request to the URL.

### Answer

A. attempts to query a DNS server on the network.

### Source explanation

By default, domain lookup function is enabled on the router so it treats every unrecognized command as a hostname, attempts to make a telnet connection to that which, in turn, attempts to resolve a hostname to IP address by querying the DNS server.
Here is a typical sample.
R1#xyz
Translating “xyz”…domain server (255.255.255.255)
Translating “xyz”…domain server (255.255.255.255) (255.255.255.255)
Translating “xyz”…domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address
Note: In order to save some time we can disable DNS lookup by the “no ip domain lookup” global command.

Source: CCNA Training » DNS Quiz.pdf | Question 1

# Question 2

What facilitates a Telnet connection between devices by entering the device name?

## Options

A. syslog

B. NTP

C. SNMP

D. DNS lookup

## Answer

D. DNS lookup

## Source explanation

Your router can be configured to use DNS lookups if we wish to use the ping or traceroute commands with a host name rather than an IP address. For example:
R1(config)#ip domain-lookup
R1(config)#ip name-server 10.10.10.1
Then we can Telnet or ping to it:
Router# ping www.cisco.com
Translating 'www.cisco.com'...domain server (10.10.10.1) [OK]
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 198.133.219.25, timeout is 2 seconds:
!!!!!
Reference: https://www.cisco.com/c/en/us/support/docs/ip/domain-name-system-dns/24182-reversedns.html
Note: In the above example, 10.10.10.1 is not the IP address of www.cisco.com. This IP address is just the DNS Name Server to resolve “www.cisco.com” to the public IP address “198.133.219.25”.
Note:
By default, any single word entered on an IOS device that is not recognized as a valid command is treated as a hostname to which you want to telnet. The device will try to translate that word to an IP address in a process that can last about a minute.
Consider the following example:
R1#test
Translating 'test'...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address
In the output above we typed an unrecognized command “test”. The router entered the DNS resolution process
which lasted about a minute. This can be annoying and this is why this feature is often turned off, especially in the lab environments.
If you don’t need to have a DNS server configured for your router, you can use the “no ip domain-lookup” command to disable the DNS translation process:
R1(config)#no ip domain-lookup
Now, if I mistype a command, the router will not perform a DNS resolution process:
R1#test
Translating 'test'
% Unknown command or computer name, or unable to find computer address
R1#

Source: CCNA Training » DNS Quiz.pdf | Question 2

# Question 3

Which command must be present in a Cisco Device configuration to enable the device to resolve an FQDN?

## Options

A. `ip domain-name`

B. `ip domain-lookup`

C. `ip name-server`

D. `ip host`

## Answer

B. ip domain-lookup

## Source explanation

FQDN stands for Fully Qualified Domain Name. The FQDN represents the absolute address of the internet presence. “Fully qualified” refers to the unique identification that guarantees that all of the domain levels are specified. The FQDN contains the host name and domain, including the top level domain, and can be uniquely assigned to an IP address. For example:
R1.9tut.com.
The command “ip domain-lookup” enables DNS-based host name-to-address translation. This command is enabled by default.

Source: CCNA Training » DNS Quiz.pdf | Question 3

# Question 4

What are two roles of Domain Name Services (DNS)? (Choose two)

## Options

A. builds a flat structure of DNS names for more efficient IP operations

B. improves security by protecting IP addresses under Fully Qualified Domain Names (FQDNs)

C. allows a single host name to be shared across more than one IP address

D. enables applications to identify resources by name instead of IP address

E. encrypts network traffic as it travels across a WAN by default

## Answer

C. allows a single host name to be shared across more than one IP address

D. enables applications to identify resources by name instead of IP address

## Source explanation

An example of DNS is described below:
When you attempt to go to a domain name such as 9tut.com, your browser will instruct your computer to do a DNS lookup on that domain name. This DNS lookup will query a DNS resolver (for example Google at 8.8.8.8). Once the resolver responds, the computer will usually choose the first IP in the response and use that for the connection. In the case of multiple IP addresses, the DNS server will respond with multiple IP addresses and the first IP will usually be used for the request. Due to how most DNS resolvers work, the order of IP addresses usually changes for each request.

Source: CCNA Training » DNS Quiz.pdf | Question 4